CSC1024 Programming Principles
Coding Exercise 9

1. Display the following menu to the user:

```
[1] Create a new file.
[2] Display the file.
[3] Add a new item to the file.
Enter 1, 2, or 3:
```

   Ask the user to enter 1, 2 or 3. If they select anything other than 1, 2 or 3 it should display a suitable error message.

   If they select 1, ask the user to enter a subject (For example, CSC1024 Programming Principles, etc.) and save it to a new file called "Subject.txt". It should overwrite any existing file with a new file.

   If they select 2, display the contents of the "Subject.txt" file.

   If they select 3, ask the user to enter a new subject and save it to the file and then display the entire contents of the file.

   Run the program several times to test the options.

2. Write a program to ask user what multiplication table the user would like to calculate. Then, store the output of the multiplication table in a text file (e.g. TT2TXT.txt). Lastly, once the multiplication table program has completed the file writing, open the file in read mode, and read the text file to display the file contents on the screen.

   Example output:

```
Display multiplication table of? 2
A multiplication table of 2 times 1 to 12.
2 x  1 =   2
2 x  2 =   4
2 x  3 =   6
2 x  4 =   8
2 x  5 =  10
2 x  6 =  12
2 x  7 =  14
2 x  8 =  16
2 x  9 =  18
2 x 10 =  20
2 x 11 =  22
2 x 12 =  24
```

3. Write a program that implements a Student Grade Manager with the following menu:

```
Student Grade Manager
[1] Add new student grades
[2] View all grades
[3] Calculate class average
```

```
[4] Exit
```

- If option 1 is selected, ask for student name and grade (0-100), then save it to "grades.txt"
- If option 2 is selected, display all grades from the file in a formatted way
- If option 3 is selected, calculate and display the class average from all grades
- If option 4 is selected, exit the program
- Handle all possible errors (file not found, invalid input, etc.)